Problem 1

Can you calculate the orbital velocity of a satellite orbiting at an altitude of 500 km?

Data: $G = 6.673 \times 10^{-11}$ SI units;

$M = 5.972 \times 10^{24}$ kg; $R = 6371000$ m;

$h = 500000$ m.

Solution:

$$v = \sqrt{\frac{6.67 \times 10^{-11} \times 5.972 \times 10^{24}}{(6371000 + 500000)}}$$

Ans: $v = 7613$ ms^{-1} or 7.613 kms^{-1}

DO YOU KNOW?

Microgravity is the condition in which people or objects appear to be weightless. The effects of microgravity can be seen when astronauts and objects float in space. Micro- means very small, so microgravity refers to the condition where gravity 'seems' to be very small.

9.4 Time period of a Satellite

Time taken by a satellite to complete one revolution round the Earth is called time period.

$$\text{Time period, } T = \frac{\text{Distance covered}}{\text{Orbital velocity}}$$

$$T = \frac{2\pi r}{v}$$

Substituting the value of v, we get

$$T = \frac{2\pi(R + h)}{\sqrt{\frac{GM}{(R + h)}}}.$$

Problem 2

At an orbital height of 500 km, find the orbital period of the satellite.

Solution:

$h = 500 \times 10^3$m, $R = 6371 \times 10^3$m,

$v = 7616 \times 10^3$ kms^{-1}.

$$T = \frac{2\pi(R + h)}{v} = 2 \times \frac{22}{7} \times \frac{(6371 + 500)}{7616}$$

$= 5.6677 \times 10^3$s $= 5667$ s.

This is $T \approx 95$ min

DO YOU KNOW?

All stars appear to us as moving from east to west, where as there is one star which appears to us stationary in its position. It has been named as Pole star. The pole star appears to us as fixed in space at the same place in the sky in the north direction because it lies on the axis of rotation of the Earth which itself is fixed and does not change its position in space. It may be noted that the pole star is not visible from the southern hemisphere.

Activity 4

Prepare a list of Indian satellites from Aryabhatta to the latest along with their purposes.

9.5 Kepler's Laws

In the early 1600s, Johannes Kepler proposed three laws of planetary motion. Kepler was able to summarize the carefully collected data of his mentor, Tycho Brahe with three statements that described the motion of planets in a Sun-centered solar system. Kepler's efforts to explain the underlying reasons for such motions are no longer accepted; nonetheless, the actual laws themselves are still considered an accurate description of the motion of any planet and any satellite. Kepler's three laws of planetary motion can be described as below.

First Law – The Law of Ellipses

All planets revolve around the Sun in elliptical orbits with Sun at one of their foci.

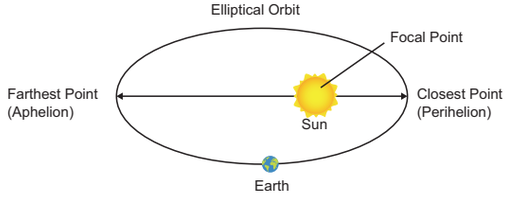


Figure 9.6 The Law of Ellipses

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